



Firehawks spread forest fires

Kites and Falcons in Australia pick up flaming sticks and drop them in other places to spread fires, and flush out prey. <http://digit.in/firehawks>



Why do teens grow quickly?

Scientists only have guesses to explain adolescent growth spurt, one of which is to age quickly to avoid trouble with adults. <http://digit.in/teenaging>

stations, where you cannot choose which tracks are to be played or skip more than 10 times per hour. The paid subscription on the same unlocks the entire library for streaming and offline playback. You can even buy individual tracks for offline playback and streaming if so desired.

Indian platforms including Gaana, Saavn, Hungama offer their entire catalogue for free with ads, a la Spotify, along with multi-device playback up to 5 devices while limiting certain features such as HD streaming and downloads. The HD is not worth it if you don't have right audio equipment to tell the difference. Wynk offers discount pricing for Airtel users, whereas Jio Music is entirely unavailable to non-Jio users.

Spotify is the most expensive at \$9.99, but the features and the recommendation engine on the platform make it worth it, and even though it is not officially available in India, people have found ways to make it work here.

Most of these platforms offer free trials for you to use and judge them first hand. Starting from Spotify's short 7-day trial, Play Music and Gaana are limited at 14 days, and Amazon Music, with the Prime trial, goes up to 30 days along with Hungama Music. Apple Music takes this one up with a three month trial period. So, if you just want to play music for free, Gaana, Saavn and Spotify are obvious choices. For specific tracks at high quality, you'll have to shell some cash, regardless of the platform.

AUDIO PLAYBACK

Music streaming services allow you to play music at different bitrates. Essentially, lower the bitrate, lower the audio quality whereas higher bitrate gives you higher audio quality. Services don't explicitly specify the

audio streaming bitrate inside the app. Instead, they are simplified to display them as low, medium and high quality. The bitrate also determines how much data is going to be used. Low space in

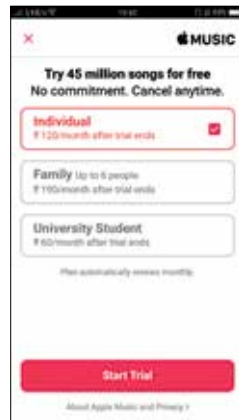
mode as well. These services aren't available in India. So, you are left with a maximum of either 128 or 160 kbps in the free version. It's also important to note that not everyone will be able to differentiate between the top tiers. However, you will notice the stark difference between the highest and lowest tier. The quality audibly distorts as you go down the tiers.

Switching between the streaming quality matters if you have limited data plans. In the higher tier of audio quality, you will lose a lot of data comparatively. Some apps do consist of an automatic mode to switch to the best quality based on the available bandwidth. If you're dealing with unlimited data then you don't need to worry.

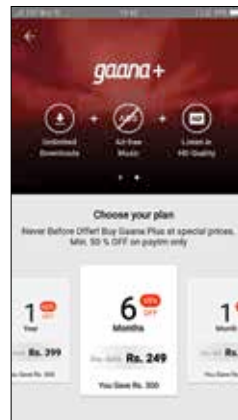
Music streaming services are protective about their bitrates. Either they don't mention it easily or it's difficult to find. From among the ones we know, Gaana offers the lowest bitrate starting from 16 kbps. It will definitely save a lot of data but the songs will noticeably sound poor. Standard quality is mostly around 64 kbps or 96 kbps. This is usually the default bitrate. If you have sharp hearing abilities, you will be able to distinguish between the two bitrates. However, as it goes higher than 128 or 160 kbps, it becomes difficult to tell them apart. The highest bitrate offered for free out of the known ones is 160 kbps. Anything higher than that falls under the paid model. For a premium service, you are given 256-320 kbps. The only services that were explicitly stated their bitrate among those compared were Gaana, Saavn, Wynk, Jio Music and Spotify.

CONTENT

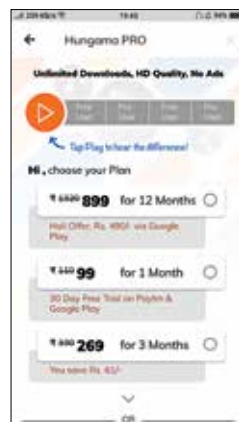
There was a time when it made sense to pick the streaming service depending on the kind of music you listen to, and the big draws were the exclusives. Streaming services are moving away from exclusives because the content is not original, and at the same time it is considered an unhealthy practice for the industry. However, the sheer size of the library is important here, as well as the composition of that library. Apart



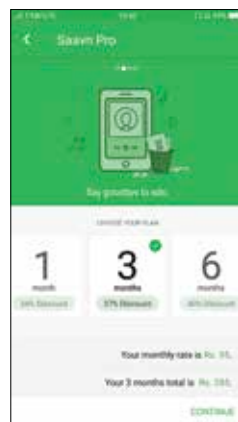
Apple Music



Gaana



Hungama



Saavn

your local storage can be a problem since apps cache the songs temporarily while streaming. This enables you to play the song again without fetching data all over again. When the app is closed those songs are removed from the cache. Play Music caches these songs permanently. They are as good as downloaded songs.

The lowest quality is usually used to save data. The bitrate ranges from 16 kbps to 96 kbps in the minimum offerings while it extends up to 320 kbps on the highest tier. These are compressed and lossy audio files. There are other services such as Tidal, Deezer, etc., that offer uncompressed and lossless audio up to bitrates of 1,411 kbps. Spotify is reportedly testing a lossless audio